C

LINEAR ALGEBRA • UNIT 7.01

Linear Transformations & Change of Basis

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 B	Q3 D	Q4 D	Q5 A
Q6 D	Q7 A	Q8 B	Q9 D	Q10 C
Q11 A	Q12 A	Q13 B	Q14 D	Q15 B
Q16 A	Q17 D	Q18 B	Q19 C	Q20 B
Q21 C	Q22 A	Q23 D	Q24 D	Q25 A

Linear Transformations & Change of Basis Practice Exercises

1 MCQ

Let $T(\mathbf{x}) = A\mathbf{x}$ with $A = \begin{bmatrix} -2 & -2 \\ -1 & 0 \end{bmatrix}$. Find $T(\langle 3, -3 \rangle)$.

- **A** $\langle 0, -3 \rangle$ ✓
- **B** $\langle -3, -6 \rangle$
- **C** $\langle 0, 3 \rangle$
- **D** $\langle 1, -2 \rangle$

WORKED SOLUTION & VERIFICATION

Multiply: $\begin{bmatrix} -2 & -2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \end{bmatrix}$.

2 MCQ

Which matrix represents the reflection across the line y = x?

- **A** $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$
- **B** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ✓
- **C** $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$
- **D** $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Find the images of $\mathbf{e}_1, \mathbf{e}_2$: $\langle 0, 1 \rangle$ and $\langle 1, 0 \rangle$. These are the columns.

3 MCQ

Which map is linear?

- **A** $T(x, y) = (x + 1,\ y)$
- **B** $T(x, y) = (xy,\ x)$
- **C** $T(x, y) = (\sin x,\ y)$
- **D** $T(A) = A^T$ on $M_{2\times 2}$ ✓

WORKED SOLUTION & VERIFICATION

$T(A) = A^T$ on $M_{2\times 2}$ preserves sums and scalar multiples. The others fail: $T(x, y) = (x + 1,\ y)$ (T(0) ≠ 0); $T(x, y) = (xy,\ x)$ (xy is not linear); $T(x, y) = (\sin x,\ y)$ (sin is not linear).

4 MCQ

Let $T(\mathbf{x}) = A\mathbf{x}$ with $A = \begin{bmatrix} -3 & -1 \\ 2 & -2 \end{bmatrix}$. Find $T(\langle -3, 2 \rangle)$.

A $\langle 13, -1 \rangle$

B $\langle -7, 10 \rangle$

C $\langle 8, -9 \rangle$

D $\langle 7, -10 \rangle$ ✓

WORKED SOLUTION & VERIFICATION

Multiply: $\begin{bmatrix} -3 & -1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ -10 \end{bmatrix}$.

5 MCQ

A 3×3 symmetric matrix T_a has eigenvalues $\lambda_1 = 20$, $\lambda_2 = 22$, and $\lambda_3 = 20$. Find the exact value of the trace $\operatorname{tr}(T_a)$.

A 62 ✓

B 40

C 82

D 65

WORKED SOLUTION & VERIFICATION

The trace equals $\lambda_1 + \lambda_2 + \lambda_3 = 62$.

6 MCQ

A linear map $f : \mathrm{R}^{20} \to \mathrm{R}^{20}$ has rank $\operatorname{rank}(f) = 18$. Determine the exact value of the nullity $\dim(\ker f)$.

A 20

B 5

C 0

D 2 ✓

WORKED SOLUTION & VERIFICATION

By Rank-Nullity, $\dim(\ker f) = 20 - 18 = 2$.

7 MCQ

Which matrix represents the orthogonal projection onto the x-axis?

A $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ ✓

B $\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$

C $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

D $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Find the images of $\mathbf{e}_1, \mathbf{e}_2$: $\langle 1, 0 \rangle$ and $\langle 0, 0 \rangle$. These are the columns.

8 MCQ

Which map is NOT linear?

A $T(x, y, z) = (x + z,\ 0)$

B $T(A) = \det A$ on $M_{2\times2}$ ✓

C $T(p) = p(1)$ on P_3

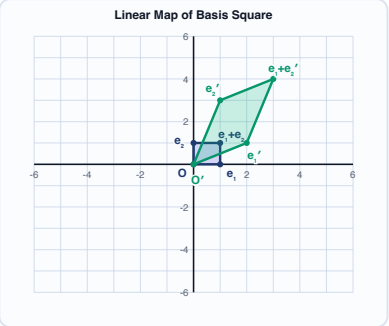
D $T(A) = A^T$ on $M_{2\times2}$

WORKED SOLUTION & VERIFICATION

$T(A) = \det A$ on $M_{2\times2}$ is not linear: det(2A) = 4det A. The other three preserve sums and scalar multiples.

9 MCQ

A linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.



A $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$

B $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

C $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$

D $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

To find $T(\mathbf{v})$, we multiply the matrix A by the vector $\mathbf{v}$. $T(\mathbf{v}) = A\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Performing the matrix-vector multiplication: $T(\mathbf{v}) = \begin{pmatrix} (2 \times 1) + (1 \times 2) \\ (1 \times 1) + (3 \times 2) \end{pmatrix} = \begin{pmatrix} 2 + 2 \\ 1 + 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$. Thus, the correct answer is $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$.

10 MCQ

Let $T : \mathrm{R}^2 \to \mathrm{R}^2$ be a linear transformation represented by the matrix $A = (2 \quad k\ 1 \quad 3)$ with respect to the standard basis. The image of the vector $\mathbf{v} = (1\ 1)$ under T is $T(\mathbf{v}) = (3\ 4)$. Find the value of the parameter k.

A 4

B 2

C 1 ✓

D 3

WORKED SOLUTION & VERIFICATION

The linear transformation T is represented by the matrix $A = (2 \quad k\ 1 \quad 3)$. The action of T on a vector $\mathbf{v}$ is given by $T(\mathbf{v}) = A\mathbf{v}$. We are given that $\mathbf{v} = (1\ 1)$ and $T(\mathbf{v}) = (3\ 4)$. Therefore, we have the matrix equation:

$$(2 \quad k\ 1 \quad 3)(1\ 1) = (3\ 4)$$

Performing the matrix multiplication on the left side gives:

$$(2(1) + k(1)\ 1(1) + 3(1)) = (2 + k\ 4)$$

Equating this to the given image vector:

$$(2 + k\ 4) = (3\ 4)$$

From the first component, we have the equation $2 + k = 3$. Solving for k, we get $k = 3 - 2 = 1$. The second component, $4 = 4$, is consistent.
Therefore, the value of k is 1.

11 MCQ

Let $T : \mathrm{R}^2 \to \mathrm{R}^2$ be a linear transformation represented by the matrix $A = (3 \quad -1\ 1 \quad 2)$ with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. Consider a new basis $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 = (1\ 1)$ and $\mathbf{v}_2 = (1\ -1)$. Let P be the change-of-basis matrix from basis B to the standard basis. Find the matrix representation of T with respect to basis B, denoted by $[T]_B$. This matrix can be expressed as $P^{-1}AP$. Find the sum of the diagonal entries of $[T]_B$.

A 5 ✓

B 6

C 4

D 7

WORKED SOLUTION & VERIFICATION

The change-of-basis matrix P from basis B to the standard basis has the vectors of B as its columns. Thus, $P = (1 \quad 1\ 1 \quad -1)$. The inverse of P is $P^{-1} = \frac{1}{\det(P)}(-1 \quad -1\ -1 \quad 1)$. Since $\det(P) = (1)(-1) - (1)(1) = -2$, we have $P^{-1} = \frac{1}{-2}(-1 \quad -1\ -1 \quad 1) = (1/2 \quad 1/2\ 1/2 \quad -1/2)$. The matrix representation of T with respect to basis B is given by $[T]_B = P^{-1}AP$. First, we compute AP: $AP = (3 \quad -1\ 1 \quad 2)(1 \quad 1\ 1 \quad -1) =$ $((3)(1) + (-1)(1) \quad (3)(1) + (-1)(-1)\ (1)(1) + (2)(1) \quad (1)(1) + (2)(-1)) = (2 \quad 4\ 3 \quad -1)$. Now, we compute $[T]_B = P^{-1}(AP)$: $[T]_B = (1/2 \quad 1/2\ 1/2 \quad -1/2)(2 \quad 4\ 3 \quad -1) =$ $((1/2)(2) + (1/2)(3) \quad (1/2)(4) + (1/2)(-1)\ (1/2)(2) + (-1/2)(3) \quad (1/2)(4) + (-1/2)(-1)) =$ $(1 + 3/2 \quad 2 - 1/2\ 1 - 3/2 \quad 2 + 1/2) = (5/2 \quad 3/2\ -1/2 \quad 5/2)$. The sum of the diagonal entries of $[T]_B$ is the trace of the matrix, which is $\frac{5}{2} + \frac{5}{2} = \frac{10}{2} = 5$.

12 MCQ

A linear map $B : \mathrm{R}^{24} \to \mathrm{R}^{24}$ has rank $\mathrm{rank}(B) = 15$. Determine the nullity $\dim(\ker B)$.

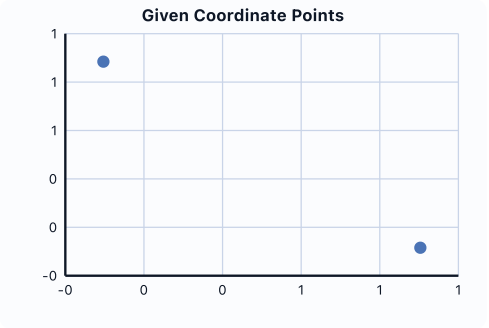
- **A** 9 ✓
- **B** 0
- **C** 12
- **D** 24

WORKED SOLUTION & VERIFICATION

By Rank-Nullity, $\dim(\ker B) = 24 - 15 = 9$.

13 MCQ

Let $T : \mathrm{R}^2 \to \mathrm{R}^2$ be a linear transformation such that $T(x, y) = (3x - y, x + 2y)$. Consider the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for R^2, where $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$. Let $\mathrm{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ be another basis for R^2, where $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -1)$. If the matrix representation of T with respect to the basis B is $[T]_{\mathrm{B}} = (a \quad b\, c \quad d)$, and $a + d = 5$, determine the value of $\det(T)$.



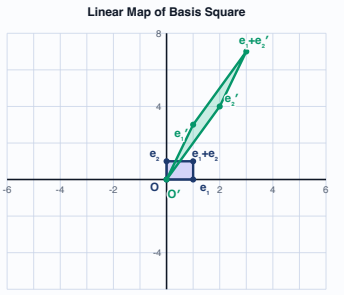
- **A** 5
- **B** 7 ✓
- **C** 3
- **D** 2

WORKED SOLUTION & VERIFICATION

The matrix representation of T with respect to the standard basis $\mathrm{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$ is $A = (3 \quad -1\ 1 \quad 2)$. The determinant of T is the determinant of its matrix representation with respect to any basis, so $\det(T) = \det(A) = (3)(2) - (-1)(1) = 6 + 1 = 7$. The information about the basis B and $[T]_{\mathrm{B}} = (a \quad b\, c \quad d)$ with $a + d = 5$ is a distractor. The trace of a matrix is invariant under a change of basis, so $\mathrm{tr}([T]_{\mathrm{B}}) = \mathrm{tr}(A)$. We have $\mathrm{tr}(A) = 3 + 2 = 5$. If $[T]_{\mathrm{B}} = (a \quad b\, c \quad d)$, then $\mathrm{tr}([T]_{\mathrm{B}}) = a + d$. The problem states $a + d = 5$, which is consistent with the trace invariance but does not directly help in finding the determinant without further calculations involving change of basis matrices. However, the determinant of a linear transformation is independent of the chosen basis. Therefore, $\det(T) = \det(A) = 7$.

14 MCQ

Consider the linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ represented by the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. If a vector $\mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is transformed by T, what is the resulting vector $T(\mathbf{v})$?



A $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$

B $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

C $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$

D $\begin{pmatrix} 4 \\ 10 \end{pmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

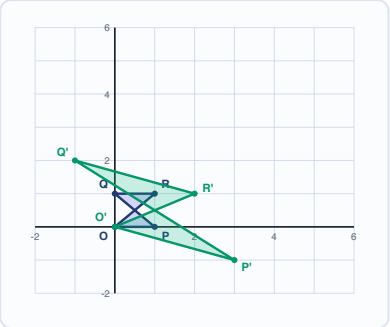
The linear transformation T can be applied to a vector $\mathbf{v}$ by multiplying the matrix A by the vector $\mathbf{v}$.

$$T(\mathbf{v}) = A\mathbf{v} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

To perform this matrix-vector multiplication, we take the dot product of each row of the matrix with the vector:
Row 1: $(1)(2) + (2)(1) = 2 + 2 = 4$
Row 2: $(3)(2) + (4)(1) = 6 + 4 = 10$
Therefore, the resulting vector is $\begin{pmatrix} 4 \\ 10 \end{pmatrix}$.

15 MCQ

A linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ is defined by $T(x, y) = (ax + by, cx + dy)$. The matrix representation of T with respect to the standard basis is $A = (a \quad b\, c \quad d)$. The image of the standard basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ under T are given by $T(\mathbf{e}_1) = (3, -1)$ and $T(\mathbf{e}_2) = (1, 2)$. The transformation is applied to a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$. The area of the transformed square is known to be k. Find the value of k.



A. 10

B. 5 ✓

C. 15

D. 20

WORKED SOLUTION & VERIFICATION

The matrix representation of the linear transformation T with respect to the standard basis is formed by the images of the standard basis vectors as its columns. Therefore, $A = (3 \quad 1\, -1 \quad 2)$. The area of the transformed region is the absolute value of the determinant of the matrix A multiplied by the area of the original region. The original region is a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$, which has an area of $1 \times 1 = 1$. The determinant of A is $\det(A) = (3)(2) - (1)(-1) = 6 - (-1) = 7$. The area of the transformed square is $|\det(A)| \times$ Area of original square $= |7| \times 1 = 7$.

16 MCQ

Let B be the basis $\langle -1, -2, 1\rangle$, $\langle 1, -1, 0\rangle$, $\langle 2, -1, 0\rangle$ of R^3. Find $[\mathbf{v}]_\mathrm{B}$ for $\mathbf{v} = \langle 1, -1, 0\rangle$.

A. $\langle 0, 1, 0\rangle$ ✓

B. $\langle 1, -1, 0\rangle$

C. $\langle -2, -1, -1\rangle$

D. $\langle 0, -1, 0\rangle$

WORKED SOLUTION & VERIFICATION

Solve $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + c_3\mathbf{b}_3 = \mathbf{v}$ (i.e. $B\mathbf{c} = \mathbf{v}$); the solution is $\langle 0, 1, 0\rangle$.

17 MCQ

Let B be the basis $\langle 2, 1, 0\rangle$, $\langle 1, 1, -1\rangle$, $\langle -2, -1, 1\rangle$ of R^3. Find $[\mathbf{v}]_\mathrm{B}$ for $\mathbf{v} = \langle 4, 3, -3\rangle$.

A. $\langle 0, -2, 1\rangle$

B. $\langle 4, 3, -3\rangle$

C. $\langle 2, 3, -3\rangle$

D. $\langle 0, 2, -1\rangle$ ✓

WORKED SOLUTION & VERIFICATION

Solve $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + c_3\mathbf{b}_3 = \mathbf{v}$ (i.e. $B\mathbf{c} = \mathbf{v}$); the solution is $\langle 0, 2, -1\rangle$.

18 MCQ

Let B be the basis $\langle -1,-1,1\rangle$, $\langle -1,-2,2\rangle$, $\langle -2,1,-2\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle 8,3,-1\rangle$.

A $\langle 8,3,-1\rangle$

B $\langle -3,-1,-2\rangle$ ✓

C $\langle 2,1,9\rangle$

D $\langle 3,1,2\rangle$

WORKED SOLUTION & VERIFICATION

Solve $c_1\mathbf{b}_1+c_2\mathbf{b}_2+c_3\mathbf{b}_3=\mathbf{v}$ (i.e. $B\mathbf{c}=\mathbf{v}$); the solution is $\langle -3,-1,-2\rangle$.

19 MCQ

Let B be the basis $\langle -1,0,-1\rangle$, $\langle -2,1,2\rangle$, $\langle -2,0,-1\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle 7,0,4\rangle$.

A $\langle 4,-4,5\rangle$

B $\langle 1,0,3\rangle$

C $\langle -1,0,-3\rangle$ ✓

D $\langle 7,0,4\rangle$

WORKED SOLUTION & VERIFICATION

Solve $c_1\mathbf{b}_1+c_2\mathbf{b}_2+c_3\mathbf{b}_3=\mathbf{v}$ (i.e. $B\mathbf{c}=\mathbf{v}$); the solution is $\langle -1,0,-3\rangle$.

20 MCQ

Let B be the basis $\langle -1,2,-1\rangle$, $\langle -2,-1,2\rangle$, $\langle 1,-1,0\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle -2,4,-2\rangle$.

A $\langle -2,0,0\rangle$

B $\langle 2,0,0\rangle$ ✓

C $\langle -2,4,-2\rangle$

D $\langle -2,-4,2\rangle$

WORKED SOLUTION & VERIFICATION

Solve $c_1\mathbf{b}_1+c_2\mathbf{b}_2+c_3\mathbf{b}_3=\mathbf{v}$ (i.e. $B\mathbf{c}=\mathbf{v}$); the solution is $\langle 2,0,0\rangle$.

21 MCQ

Let B be the basis $\langle 0,-1,1\rangle$, $\langle -1,-1,0\rangle$, $\langle 1,0,2\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle -2,-5,3\rangle$.

A $\langle -2,-5,3\rangle$

B $\langle -3,-2,0\rangle$

C $\langle 3,2,0\rangle$ ✓

D $\langle 4,2,0\rangle$

WORKED SOLUTION & VERIFICATION

Solve $c_1\mathbf{b}_1+c_2\mathbf{b}_2+c_3\mathbf{b}_3=\mathbf{v}$ (i.e. $B\mathbf{c}=\mathbf{v}$); the solution is $\langle 3,2,0\rangle$.

22 MCQ

Let B be the basis $\langle 0,-1,1\rangle$, $\langle -1,-1,1\rangle$, $\langle -2,2,-1\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle -1,4,-3\rangle$.

A $\langle -1,-1,1\rangle$ ✓

B $\langle -1,4,-3\rangle$

C $\langle 2,3,-1\rangle$

D $\langle 1,1,-1\rangle$

WORKED SOLUTION & VERIFICATION

Solve $c_1\mathbf{b}_1+c_2\mathbf{b}_2+c_3\mathbf{b}_3=\mathbf{v}$ (i.e. $B\mathbf{c}=\mathbf{v}$); the solution is $\langle -1,-1,1\rangle$.

23 MCQ

Let B be the basis $\langle 1,-1,-1\rangle$, $\langle 0,2,-1\rangle$, $\langle -1,0,1\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle 6,-5,-5\rangle$.

- A $\langle 6,-5,-5\rangle$
- B $\langle 7,1,-6\rangle$
- C $\langle -3,1,3\rangle$
- D $\langle 3,-1,-3\rangle$ ✓

WORKED SOLUTION & VERIFICATION

Solve $c_1\mathbf{b}_1+c_2\mathbf{b}_2+c_3\mathbf{b}_3=\mathbf{v}$ (i.e. $B\mathbf{c}=\mathbf{v}$); the solution is $\langle 3,-1,-3\rangle$.

24 MCQ

Let B be the basis $\langle -1,1,1\rangle$, $\langle 1,0,0\rangle$, $\langle -1,2,1\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle 0,-6,-3\rangle$.

- A $\langle 0,-6,-3\rangle$
- B $\langle -6,0,-9\rangle$
- C $\langle 0,3,3\rangle$
- D $\langle 0,-3,-3\rangle$ ✓

WORKED SOLUTION & VERIFICATION

Solve $c_1\mathbf{b}_1+c_2\mathbf{b}_2+c_3\mathbf{b}_3=\mathbf{v}$ (i.e. $B\mathbf{c}=\mathbf{v}$); the solution is $\langle 0,-3,-3\rangle$.

25 MCQ

Let B be the basis $\langle -2,1,2\rangle$, $\langle -1,2,0\rangle$, $\langle 0,1,-1\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle 2,-7,3\rangle$.

- A $\langle 0,-2,-3\rangle$ ✓
- B $\langle -8,-4,1\rangle$
- C $\langle 0,2,3\rangle$
- D $\langle 2,-7,3\rangle$

WORKED SOLUTION & VERIFICATION

Solve $c_1\mathbf{b}_1+c_2\mathbf{b}_2+c_3\mathbf{b}_3=\mathbf{v}$ (i.e. $B\mathbf{c}=\mathbf{v}$); the solution is $\langle 0,-2,-3\rangle$.